Portland State UNIVERSITY

APPLIED PROMPT ENGINEERING ASSIGNMENT GUIDE

The Scenario: You've just been hired as a Business Analyst Intern at a mid-sized SaaS company ($50M in annual revenue) that builds a language learning app targeted at English speakers learning Spanish, French, and Japanese. The app offers short daily lessons, vocabulary quizzes, and practice games. The company's leadership is interested in increasing user retention, especially among new users, in the first 30 days.

Your manager asks you to use a generative AI tool (like ChatGPT) to research what top competitors are doing to keep users engaged (product features and behavioral design).

They want to know:

- What gamification strategies are working for other language learning apps?
- How are successful apps using notifications, reminders, or social components to bring users back?
- Are there any lessons from social media or other addictive apps that could be ethically applied to increase retention and increase engagement, without compromising trust?

Unfortunately, your first prompt returned vague results. Your task is to revise and improve it, step by step, to generate more actionable insights.

Assignment Overview

You will start with a basic prompt and improve it in 3 rounds of revision, documenting your thought process, the changes you make, and reflecting on what you notice. You will be graded on prompt development, the quality of your output improvement, and the quality and depth of your reflection.

Your Goal: Learn how to write and iterate on AI prompts to generate useful business information.

Part 1: Baseline Prompt

- **Use** this basic prompt:
  - "What are other companies doing with language learning apps?"
- **Copy/paste** the output and briefly evaluate

    - What worked in this prompt?
    - What was too vague, missing, or irrelevant
    - Was there too much information, too little, just right?
- **Write your response in 3–4 complete sentences.** Keep your reflections concise but specific. You'll follow this format for each prompt revision throughout the assignment.

Part 2: The First Revision

- **Revise** the original prompt using these enhancements
    - Provide context (*e.g. for a mid-size company developing a language learning app)*
    - Specify the type of learning method or feature (*e.g., gamified lessons, streak tracking, or AI tutors*)
    - Consider adding a response limit (*e.g., "in ____ (#) of bullet points" or "keep it under ____ (#) of words")*
        - Be careful not to limit the response too much. You don't want to lose important nuance or detail.
- Remember that your company is trying to **increase retention** and learn from competitors in the **language learning app space**. Focus your prompt on gathering ideas that a product or marketing team could act on.

THEN:

- **Copy/paste** the new output and reflect on what changed
- **Discuss** if the output improved. Was any detail lost because of your character limits?
- **Write your response in 3–4 complete sentences.**

Part 3: The Second Revision

- **Add** another layer of context and specificity
    - Ask for **specific examples**, **trends**, **company names, or anything else you feel is relevant.**
    - Add another layer of context or specificity (e.g., "targeting ____ learners using mobile devices")
    - Ask for specific examples, trends, or company names related to the language learning industry
    - Tell the AI how you want it to organize the outputs
    - Add at least 3 additional prompt engineering techniques we discussed in class
        - Take on a role
        - Give ChatGPT an example of what you're looking for
        - Break the prompt into steps (e.g., "*First list competitors in mobile language learning, then summarize their engagement strategies*")
            - Insert your own text where the bold and bracketed text is.

        - Specify the output format
        - Ask the model to reflect on its output or justify its response
- Revisit the original business problem: your manager wants to know what competitors are doing to retain users and whether tactics from social media or gamification can be ethically applied. Your prompt should aim to uncover **specific insights** that could help your company design better features, messaging, or strategies.

THEN:

- **Evaluate** any improvements in relevance and usefulness
- **Reflect** on how you applied these techniques and what impact they had.
- **Write your responses in 3–4 complete sentences.**
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Part 4: The Final Prompt

- **Write** the final prompt using a combination of what you're already created and anything else you've learned about prompt engineering.
- Prompts should be:
    - Focused and well structured
    - Business professional and grammatically correct
    - Realistic for business use
    - Packed with as much context and information as contributes to the goal.
- **Tip:** Revisit the original scenario and include any useful details, such as the company's goal or to explore ethical concerns.
- **Copy/paste** the output
- **Note:** If your final prompt includes multiple steps, it may result in a short back-and-forth or conversational exchange with the model. That's expected. Include both parts of the interaction if it helps demonstrate the prompt's effectiveness.

Part 5: Final Project Reflection

- **Consider and reflect** on the following topics
    - How did the output evolve and where did you see the biggest differences?
    - Looking at your final AI prompt, which prompting technique do you think played the most important role in developing quality? (*e.g., Chain of Thought, Zero-Shot, Tree of Thought, Few-Shot, etc.*) Briefly explain your choice.
    - If you're currently working, describe a task or responsibility in your role where generative AI **could be useful**, now or in the near future. How could it assist, speed up, or improve your work? Could work quality be reduced somehow?
    - If you're not currently working, think about a job you want to have. Do you believe generative AI could play a role in that kind of work?
        - If yes: What task(s) could it help with?

        - If no: Do you think that will change as AI tools evolve? Why or why not?
- Limit each response to 1–3 sentences

Submission

- **Submit** the following
    - The 4 prompts (original + iterations)
    - Outputs
    - Individual prompt reflections
    - Final reflection
- Use the provided **Reflection Worksheet** (available in the Project Instructions Module) to format and organize your responses. This is optional but strongly recommended to keep your submission clean and complete.